

Data Scientist

<https://www.linkedin.com/in/varun-kumar-967599194/>

Preferred Location: Noida, Gurgaon, Bangalore, Hyderabad

Mobile: 8737033969

EXECUTIVE SUMMARY

Data Scientist with **4 years** of hands-on experience building scalable, ethical, and intelligent AI systems, including LLM-powered assistants, Responsible AI platforms, and real-time detection frameworks. Expert in **RAG, Agentic AI, and Federated Learning**, with strong proficiency in Python, **PySpark**, Azure, and **LangChain**. Ex-Founder and Ex-Head of Product at ScrutinAI Pvt. Ltd, responsible for leading the development of a Responsible AI SaaS platform incubated via the Mercedes-Benz Intrapreneurship Program. **Filed 4 patents** and delivered multiple award-winning enterprise AI solutions focused on fairness, explainability, AI security, and privacy. Seeking to drive impact through innovation, governance, and technically sound AI development in high-performance teams.

EDUCATION

Indian Institute of Technology, Palakkad (IIT, PKD) • <i>Master of Technology in Data Science</i>	Palakkad, India 09/2020 – 07/2022
JSS Academy of Technical Education, Noida • <i>Bachelor of Technology in Electronics and Communication</i>	Noida, India 08/2015 – 06/2019

TECHNICAL SKILLS

- **Languages:** Python, SQL, PySpark.
- **AI/ML:** Deep Learning, ML, Federated Learning, Generative AI, LLMs, RAG, CV, NLP
- **Libraries:** LangChain, TensorFlow, PyTorch, Keras, scikit-learn, pandas, NumPy, matplotlib
- **Tools:** Azure (DataBricks, Data Lake, Delta Lake, ADF), Docker, Kubernetes, FastAPI, Git, Streamlit

EXPERIENCE

Data Scientist • <i>Full Time</i>	Mercedes Benz Research and Development India 08/2023 – Present
◦ Built Responsible AI toolkit addressing fairness, privacy, and governance—cutting audit effort by 85%. ◦ Built a Responsible AI assistant that significantly reduced AI compliance and AI governance risks, ensuring robust oversight and ethical alignment of AI systems. ◦ Developed Agentic RAG-based legal assistant with Chain of thought prompting, reducing manual effort by 90% ◦ Automated ECU compliance analysis from MATLAB PDFs, reducing review time by 95%. ◦ Led real-time malware detection at EV stations using Spark + Azure, reducing threats by 80%. ◦ Founded Responsible AI SaaS platform incubated via Mercedes-Benz Intrapreneurship Program. ◦ Led development of compliance SaaS Product for fairness, explainability, AI security and privacy in ML systems.	
Data Scientist Intern • <i>Internship</i>	Mercedes Benz Research and Development India 01/2023 – 07/2023
◦ Developed advanced algorithms addressing fairness, privacy, explainability, security, and federated learning to ensure Responsible AI compliance.	
Machine Learning Engineer • <i>Full Time</i>	NeuranceAI Pvt. Ltd 11/2022 – 01/2023
◦ Improved early risk detection and treatment personalization in oncology and nephrology, reducing 90-day mortality risk and enabling proactive clinical decisions.	
Machine Learning Engineer • <i>Full Time</i>	Voiceano Voice Solutions Pvt Ltd 07/2019 – 08/2020
◦ Built intelligent customer support chatbots using QA similarity, reducing query resolution time and improving response consistency through feedback-driven training.	

PROJECTS

- **Responsible AI SaaS Product** (Mercedes Benz R & D India Project)
 - **AI Ethics | Explainability | Fairness | MLOps Integration | API Development**
 - Delivered an end-to-end Responsible AI framework addressing fairness, privacy, AI security and explainability across TensorFlow, PyTorch, and Scikit-learn models.
 - Developed APIs for seamless integration into MLOps pipelines (on-prem & cloud), supporting secure and compliant AI lifecycle workflows.
- **LLM-Govern: AI Compliance Assistant** (Mercedes Benz R& D India Project)
 - **Responsible AI | GDPR | RAG | LLAMA | LangChain**
 - Engineered a compliance assistant for GDPR-aligned AI model evaluations using a fine-tuned LLAMA model with RAG.
 - Built a web-based tool where developers input model/dataset metadata to receive compliance summaries and actionable guidance via Responsible AI tool links.
- **LexiQ-RAG: Legal Intelligence Assistant** (Mercedes Benz R& D India Project)
 - **LLM | RAG | LangChain | LLAMA 3 | Prompt Engineering**
 - Built an Agentic Retrieval-Augmented Generation (RAG) system using LLAMA 3 to streamline legal policy search across internal documents.
 - Designed conversational workflows with Chain-of-Thought prompting for interpretable reasoning.
 - Integrated human-in-the-loop validation with legal experts and fine-tuned the model using proprietary legal corpora, significantly reducing manual legal consultations.
- **CompliCheck-RAG: Functional Compliance & Defeat Detection** (Mercedes Benz R& D India Project)
 - **RAG | GPT-4 | LangChain | PDF Parsing | JSON Structuring | Simulink Analysis**
 - Developed an intelligent assistant to analyze ECU configurations from MATLAB-generated PDFs by converting diagrams into structured JSON.
 - Automated detection of defeat device behavior by mapping functional requirements to ECU parameters, delivering detailed compliance reports.
 - Enabled engineers to query functional flows, improving defect detection and regulatory alignment.
- **Real-Time Malware Detection at EV Charging Stations** (Mercedes Benz R& D India Project)
 - **Azure Databricks | PySpark | Power BI | Kafka | MLlib | Spark Streaming**
 - Built a real-time malware anomaly detection pipeline using sensor data collected during EV charging.
 - Processed streaming data using PySpark, stored results in Azure Data Lake, and visualized insights via Power BI dashboards.
 - Implemented ML models to detect and alert on malicious socket behavior, enhancing EV cybersecurity.
- **Secure Federated Learning Architecture** (MTech Thesis)
 - **Federated Learning | TensorFlow | Docker | Socket Programming | Privacy-Preserving AI**
 - Proposed a privacy-preserved Federated Learning architecture to defend against poisoning and inference attacks.
 - Implemented secure communication protocols and differential privacy techniques for decentralized model training across edge nodes.

ACHIEVEMENTS

- Winner of **Mercedes Benz Intrapreneurship Award 2025**.
- **Completed 1-year PG Diploma in Quantum Machine Learning from IIT Madras**, specializing in the convergence of quantum computing and AI.
- Successfully selected in the prestigious **Mercedes-Benz Intrapreneurship Program (MBIIP 2023)**, with our idea standing out as the **only startup idea** among 240 Proof of Concepts (POCs) in 2023 to progress to the startup stage.
- **Filed 4 patents in Mercedes Benz R&D** related to Responsible AI research, addressing key challenges in ethics, compliance, and transparency in AI systems.
- Played a pivotal role in the **end-to-end development** of the project, transforming it into a Minimum Viable Product (MVP) and positioning it as the **second-ever startup** to emerge from Mercedes-Benz India, following the first startup launched in 2021.